

实验6.2

TCP协议基本分析

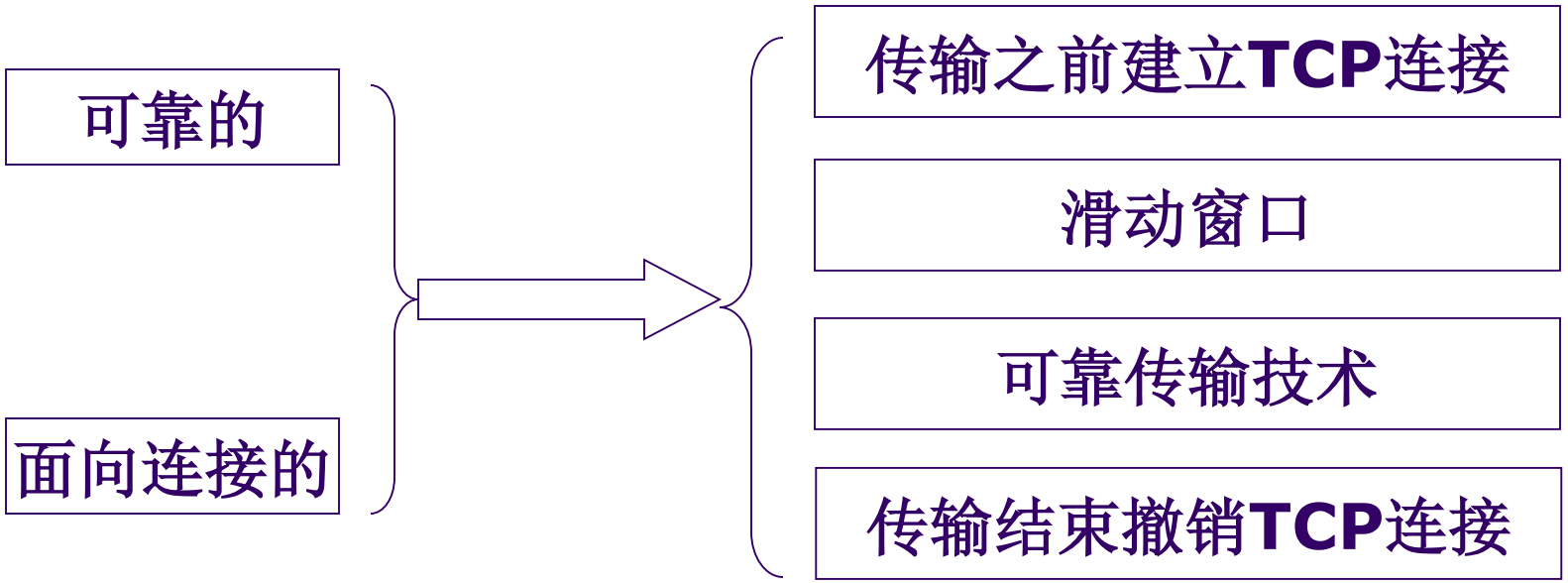
TCP协议基本分析

- TCP报文分析
- TCP连接建立
- TCP连接撤销

实验原理 — TCP协议简介

TCP协议 — Transfer Control Protocol

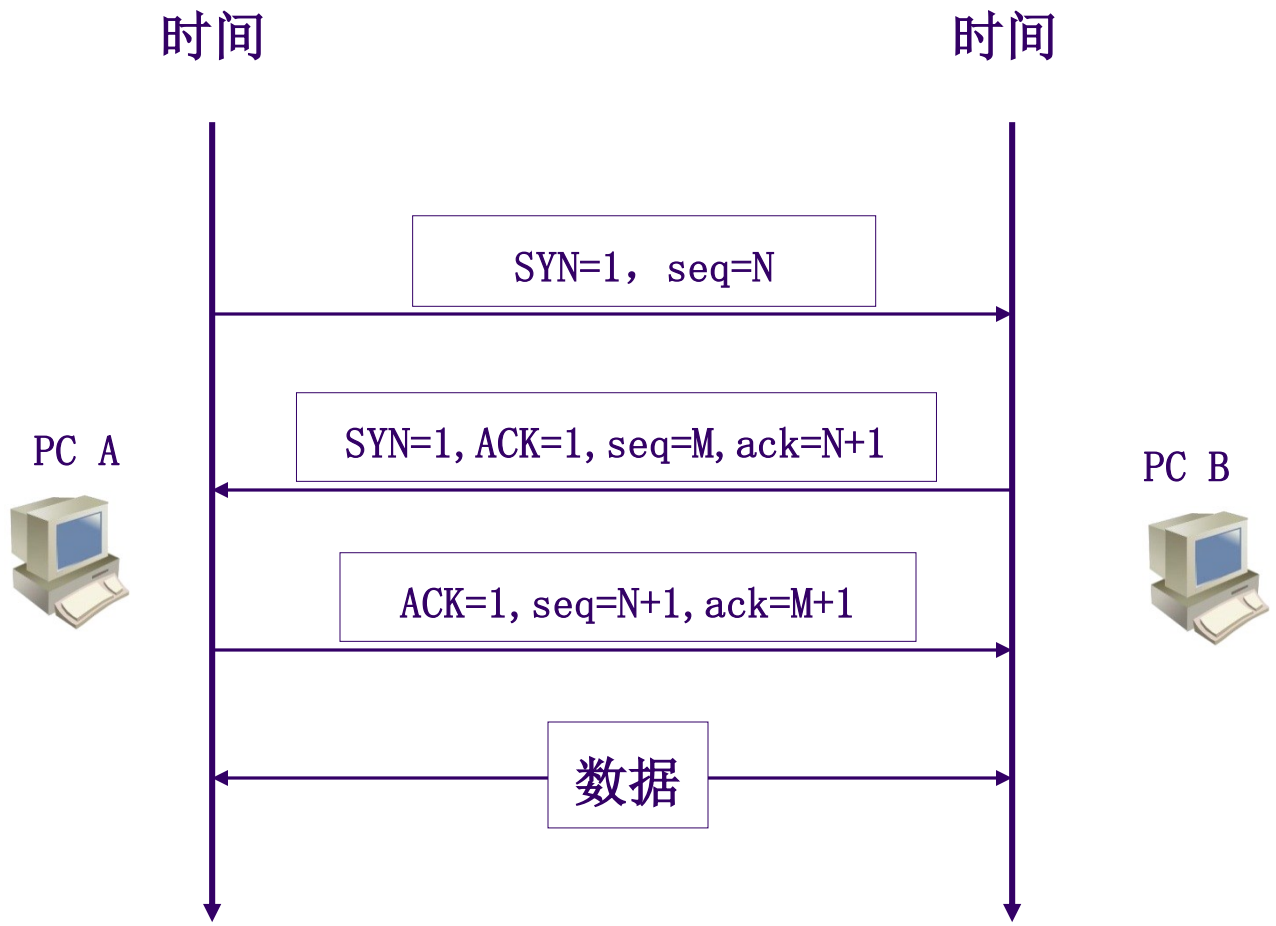
特点：



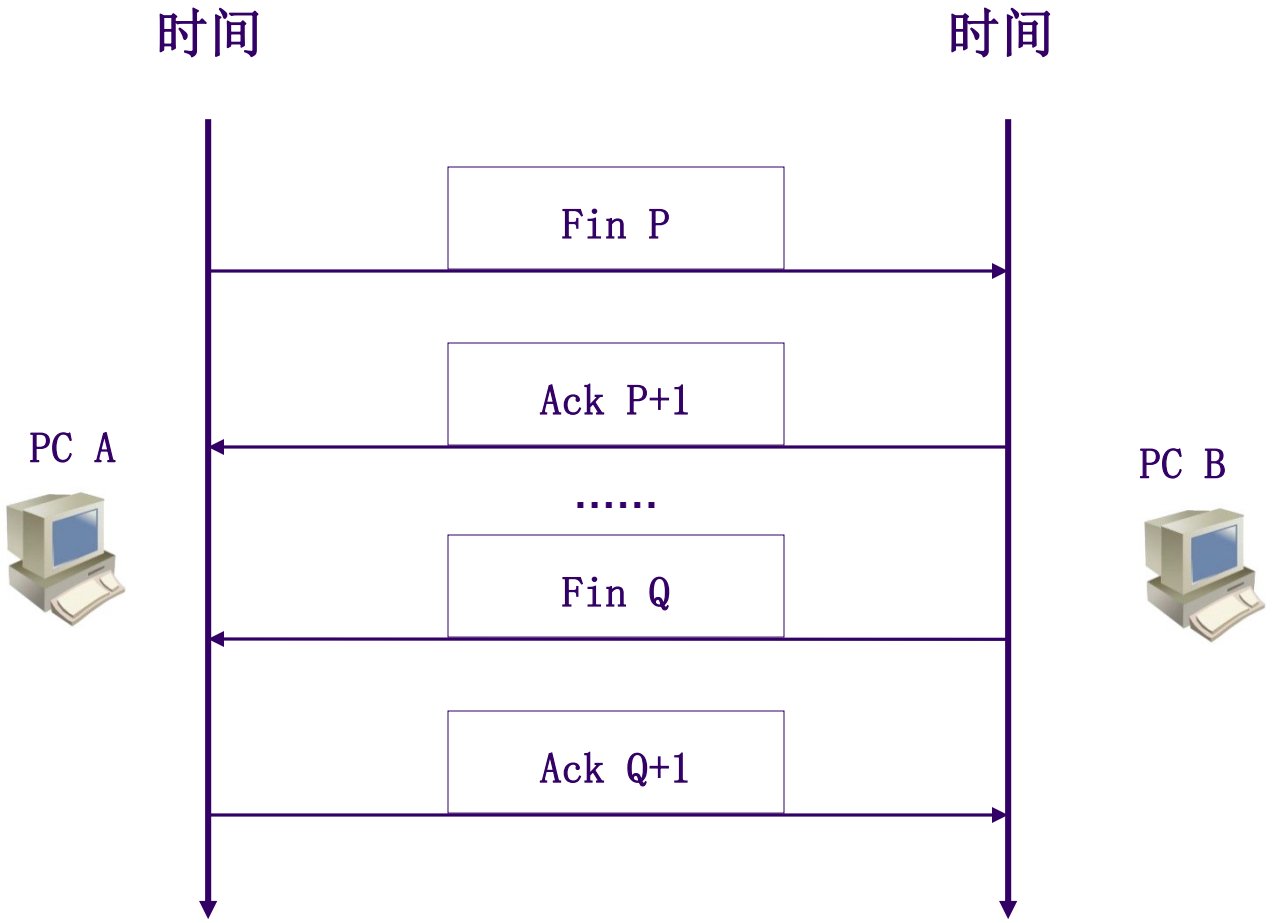
实验原理 — TCP报文格式

<table>
<tr><td colspan="4">16位源端口号</td><td colspan="4">16位目的端口号</td></tr>
<tr><td colspan="8">32位顺序号</td></tr>
<tr><td colspan="8">32位确认号</td></tr>
<tr><td>……</td><td>URG</td><td>ACK</td><td>PSH</td><td>RST</td><td>SYN</td><td>FIN</td><td>16位窗口大小</td></tr>
<tr><td colspan="5">16位校验和</td><td colspan="3">16位紧急指针</td></tr>
<tr><td colspan="8">选项</td></tr>
<tr><td colspan="8">数据</td></tr>
</table>

实验原理 — TCP建立连接过程



实验原理 — TCP撤销连接过程



注意事项

- **TCP连接建立采用什么方式？**
- 三次握手时**option**字段中的maximum segment size (*MSS*) =1460是如何计算出来的？
  - 以太网最大帧长度（**1518**）
  - 帧头和帧尾（**18**）
  - **IP**首部（**>=20**）、**TCP**首部（**>=20**）
- **TCP**报文中没有数据长度字段， **Wireshark**软件解析报文时显示的数据长度的值是如何计算出来的？给出计算公式。

option字段中的maximum segment size (*MSS*)

No.	Time	Source	Destination	Protocol	Info
1	0.000000	HuaweiTe_09:bc:	Spanning-tree-(	0x88a	Ethernet II
2	60.001867	HuaweiTe_09:bc:	Spanning-tree-(	0x88a	Ethernet II
3	74.229476	192.168.1.2	192.168.2.2	TCP	46576 > search-agent [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM=1
4	74.229517	192.168.2.2	192.168.1.2	TCP	search-agent > 46576 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1460 SACK_PERM=1
5	74.230371	192.168.1.2	192.168.2.2	TCP	46576 > search-agent [ACK] Seq=1 Ack=1 Win=5840 Len=0
6	74.231843	192.168.1.2	192.168.2.2	TCP	46576 > search-agent [PSH, ACK] Seq=1 Ack=1 Win=5840 Len=1400
7	74.231852	192.168.1.2	192.168.2.2	TCP	46576 > search-agent [ACK] Seq=1401 Ack=1 Win=5840 Len=1460
8	74.231862	192.168.2.2	192.168.1.2	TCP	search-agent > 46576 [ACK] Seq=1 Ack=1401 Win=4440 Len=0
9	74.231866	192.168.1.2	192.168.2.2	TCP	46576 > search-agent [ACK] Seq=2861 Ack=1 Win=5840 Len=1460
10	74.231928	192.168.2.2	192.168.1.2	TCP	search-agent > 46576 [ACK] Seq=1 Ack=4321 Win=1520 Len=0
11	74.233665	192.168.1.2	192.168.2.2	TCP	46576 > search-agent [ACK] Seq=4321 Ack=1 Win=5840 Len=1460
12	74.233742	192.168.2.2	192.168.1.2	TCP	search-agent > 46576 [ACK] Seq=1 Ack=5781 Win=1520 Len=0

```
⊞ Frame 4: 62 bytes on wire (496 bits), 62 bytes captured (496 bits)
⊞ Ethernet II, Src: ec:a8:6b:a0:57:2f (ec:a8:6b:a0:57:2f), Dst: HuaweiTe_48:f5:d1 (00:e0:fc:48:f5:d1)
⊞ Internet Protocol, Src: 192.168.2.2 (192.168.2.2), Dst: 192.168.1.2 (192.168.1.2)
⊟ Transmission Control Protocol, Src Port: search-agent (1234), Dst Port: 46576 (46576), Seq: 0, Ack: 1, Len: 0
    Source port: search-agent (1234)
    Destination port: 46576 (46576)
    [Stream index: 0]
    Sequence number: 0    (relative sequence number)
    Acknowledgement number: 1    (relative ack number)
    Header length: 28 bytes
  ⊞ Flags: 0x12 (SYN, ACK)
    Window size: 5840
  ⊞ Checksum: 0x3853 [validation disabled]
  ⊟ Options: (8 bytes)
      Maximum segment size: 1460 bytes
      NOP
      NOP
      TCP SACK Permitted Option: True
  ⊞ [SEQ/ACK analysis]
```

关于TCP报文的数据部分长度

No.	Time	Source	Destination	Protocol	Info
3	74.229476	192.168.1.2	192.168.2.2	TCP	46576 > search-agent [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM=1
4	74.229517	192.168.2.2	192.168.1.2	TCP	search-agent > 46576 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1460 SACK_PERM=1
5	74.230371	192.168.1.2	192.168.2.2	TCP	46576 > search-agent [ACK] Seq=1 Ack=1 Win=5840 Len=0
6	74.231843	192.168.1.2	192.168.2.2	TCP	46576 > search-agent [PSH, ACK] Seq=1 Ack=1 Win=5840 Len=1400

```
⊞ Frame 6: 1454 bytes on wire (11632 bits), 1454 bytes captured (11632 bits)
⊞ Ethernet II, Src: HuaweiTe_48:f5:d1 (00:e0:fc:48:f5:d1), Dst: ec:a8:6b:a0:57:2f (ec:a8:6b:a0:57:2f)
⊟ Internet Protocol, Src: 192.168.1.2 (192.168.1.2), Dst: 192.168.2.2 (192.168.2.2)
    Version: 4
    Header length: 20 bytes
  ⊞ Differentiated Services Field: 0x00 (DSCP 0x00: Default; ECN: 0x00)
    Total Length: 1440
    Identification: 0x191e (6430)
  ⊞ Flags: 0x02 (Don't Fragment)
    Fragment offset: 0
    Time to live: 63
    Protocol: TCP (6)
  ⊞ Header checksum: 0x98e5 [correct]
    Source: 192.168.1.2 (192.168.1.2)
    Destination: 192.168.2.2 (192.168.2.2)
⊟ Transmission Control Protocol, Src Port: 46576 (46576), Dst Port: search-agent (1234), Seq: 1, Ack: 1, Len: 1400
    Source port: 46576 (46576)
    Destination port: search-agent (1234)
    [Stream index: 0]
    Sequence number: 1    (relative sequence number)
    [Next sequence number: 1401    (relative sequence number)]
    Acknowledgement number: 1    (relative ack number)
    Header length: 20 bytes
  ⊞ Flags: 0x18 (PSH, ACK)
    Window size: 5840
  ⊞ Checksum: 0xf31e [validation disabled]
  ⊞ [SEQ/ACK analysis]
⊞ Data (1400 bytes)
```

```
0010  05 a0 19 1e 40 00 3f 06  98 e5 c0 a8 01 02 c0 a8   ....@.?. ........
0020  02 02 b5 f0 04 d2 78 e9  55 f0 64 78 c1 83 50 18   ......x. U.dx..P.
0030  16 d0 f3 1e 00 00 0d 0a  0d 0a 52 46 43 3a 20 37   ........ ..RFC: 7
0040  39 33 0d 0a 20 20 20 20  20 20 20 20 20 20 20 20   93..
```